A2 Notes 6.1 SIMPLIFYING RADICALS

$\sqrt{16}$ *is called the square root of* 16, *and it means* what # when squared = 16

nothing there means a 2 is there

$\sqrt{16} = 4$ b/c $4^2 = 16$

index → $\sqrt{\quad}$ ← The radical sign

← the radicand

Your goal in simplifying radicals is to bring out from under √ anything you can.

It is useful to recognize the following numbers:

Perfect Squares	Perfect Cubes	Perfect 4ths	Perfect 5ths
$\sqrt{1} = 1$	$\sqrt[3]{1} = 1$	$\sqrt[4]{1} = 1$	$\sqrt[5]{1} = 1$
$\sqrt{4} = 2$ b/c $2^2 = 4$	$\sqrt[3]{8} = 2$ b/c $2^3 = 8$	$\sqrt[4]{16} = 2$	$\sqrt[5]{32} = 2$
$\sqrt{9} = 3$	$\sqrt[3]{27} = 3$	$\sqrt[4]{81} = 3$ b/c $3^4 = 81$	
$\sqrt{16} = 4$	$\sqrt[3]{64} = 4$		
$\sqrt{25} = 5$	$\sqrt[3]{125} = 5$		
$\sqrt{36} = 6$	$\sqrt[3]{216} = 6$		
$\sqrt{49} = 7$			
$\sqrt{64} = 8$			
$\sqrt{81} = 9$			
$\sqrt{100} = 10$			
$\sqrt{121} = 11$			
$\sqrt{144} = 12$			

Simplify:

$\sqrt{121}$	$\sqrt[3]{125}$	$\sqrt[4]{81}$	$\sqrt[5]{32}$
11	5	3	2

BUT, not all radicals will simplify "perfectly".

To simplify radicals:

1) make a prime factorization tree by dividing the radicand by 2 until you can't, then 3 until you can't, then 5 etc.

   remember, prime numbers: 2, 3, 5, 7, 11... only have factors of 1 and itself

2) group each kind of factor into what you need:
   If square root, you need to group by pairs
   If cube root, you need to group by threes
   If fourth root, you need to group 4 things

3) pull out from under the radical sign any factor that has the correct grouping

Examples: Rewrite in simplest radical form. you!

$\sqrt{48}$ 2 · 24 2 · 12 2 · 6 2 · 3 2×2 $\sqrt{48} = \sqrt[2]{2\cdot2\cdot2\cdot2\cdot3}$ $4\sqrt{3}$	$\sqrt[3]{54}$ 2 · 27 3 · 9 3 · 3 3 $1\sqrt[3]{2\cdot3\cdot3\cdot3}$ ↑ coefficient $3\sqrt[3]{2}$	$\sqrt[4]{32}$ 2 · 16 2 · 8 2 · 4 2 · 2 2 $\sqrt[4]{2\cdot2\cdot2\cdot2\cdot2}$ $2\sqrt[4]{2}$	$\sqrt[5]{486}$ 2 · 243 3 · 81 3 · 27 3 · 9 3 · 3 3 $\sqrt[5]{2\cdot3\cdot3\cdot3\cdot3\cdot3}$ $3\sqrt[5]{2}$

Now, **what if the radical already has a coefficient...no biggie**. Multiply that coefficient by what you take out.

Examples: Rewrite in simplest radical form.

$7\sqrt{50}$ 2 · 25 5 · 5 × 5 $7\sqrt[2]{2\cdot5\cdot5}$ $35\sqrt{2}$	you $-4\sqrt{27}$ 3 · 9 3 · 3 3 $-4\sqrt[2]{3\cdot3\cdot3}$ $-12\sqrt[2]{3}$	$5\sqrt[3]{81}$ 3 · 27 3 · 9 3 · 3 3 $5\sqrt[3]{3\cdot3\cdot3\cdot3}$ $15\sqrt[3]{3}$	$-10\sqrt[3]{16}$ 2 · 8 2 · 4 2 · 2 2 $-10\sqrt[3]{2\cdot2\cdot2\cdot2}$ $-20\sqrt[3]{2}$

$$x^a \cdot x^b = x^{a+b}$$

VARIABLES

Simplifying radicals with variables is actually easier than numbers. This is due to the following:

$$\sqrt[r]{x^p} = x^{\frac{p}{r}} \quad \begin{matrix}\leftarrow \text{power} \\ \leftarrow \text{root}\end{matrix}$$

ex $\sqrt[3]{x^5} = x^{\frac{5}{3}}$ ← 5 ← divided by 3

So, **if the power is divisible by the index** you are home free! "evenly"

$x^{\text{even}/2}$ $\quad x^{67} = x^{66/2} \cdot x^1$

$x^{\text{odd}} = x^{\text{even}} \cdot x^1$

Examples: simplify completely.

$\sqrt[2]{\cancel{75}x^{10}}$ $10 \div 2$ power root Rewrite $\sqrt{x^5 \cdot x^5}$ x^5	$\sqrt[\cancel{3}\,2]{\cancel{162}x^4y^8}$ $\sqrt[2]{x^{4/2}\,y^{8/2}}$ $\boxed{x^2y^4}$	$-2\sqrt[\cancel{3}\,2]{24a^6b^9}$ $24 \to 12 \cdot 2 \to 6 \cdot 2 \to 2 \cdot 3$; $a^{6/2}$; $b^{8/2} \cdot b^1$ $-2\sqrt{2 \cdot 2 \cdot 2 \cdot 3 \cdot a^{6/2}\, b^{8/2} \cdot b^1}$ $-2 \cdot 2a^3b^4\sqrt{6b}$ $= \boxed{-4a^3b^4\sqrt{6b}}$	$\cancel{\sqrt[4]{625a^4b^{12}c^{20}}}$ $\sqrt[2]{a^4b^7c^{19}}$ $a^{4/2}\; b^{6/2} \cdot b^1\; c^{18/2} \cdot c^1$ $\boxed{a^2b^3c^9\sqrt{bc}}$

If the power is NOT divisible by the index, we need to separate the variable that is raised to the power into two factors:

2 easier ones on back before these hard ones

One that is divisible by n, and the other with the "remainder power" (leftover)

Simplify completely.

$5x^5\sqrt{72x^7}$ $72 \to 36 \cdot 2 \to 6 \cdot 6$; $x^{6/2} \cdot x^1$ $5x^5 \cdot 6 \cdot x^3\sqrt{2x}$ $\boxed{30x^8\sqrt{2x}}$	$xy\sqrt{98x^3y^{13}}$ $98 \to 2 \cdot 49 \to 7 \cdot 7$ $xy\sqrt[2]{2 \cdot 7 \cdot 7 \cdot x^{2/2} \cdot x^1 \cdot y^{12/2} \cdot y^1}$ $x^1 \cdot y^1 \cdot 7 \cdot x^1 \cdot y^6\sqrt{2xy}$ $7x^2y^7\sqrt{2xy}$	$8y^2\sqrt[3]{24x^7y^{11}}$ $\cancel{8y^2\sqrt[3]{2 \cdot 2 \cdot 2}}$ $8y^2\sqrt[3]{24x^7y^{11}}$ $24 \to 2 \cdot 12 \to 2 \cdot 6 \to 2 \cdot 3$ $8y^2\sqrt[3]{2 \cdot 2 \cdot 2 \cdot 3 \cdot x^6 \cdot x^1 \cdot y^9 \cdot y^2}$ $2x^2 \cdot y^3$	$\sqrt[4]{32a^5b^{10}c^{15}}$ $32 \to 2 \cdot 16 \to 2 \cdot 8 \to 2 \cdot 4 \to 2 \cdot 2$; $a^4 \cdot a$; $b^{8/4} \cdot b^2$; $c^{12/4} \cdot c^3$ $2ab^2c^3\sqrt[4]{2abc^3}$ (crossed out: $2ab^2c^3\sqrt[4]{2b^2c^3}$) $2ab^2c^3\sqrt[4]{2ab^2c^3}$

So, 1) prime factorization of any # under the radical, pull out appropriate group

2) pull out what you can of the variable

$$8y^2\sqrt[3]{3 \cdot x\, y^2}$$

$$\boxed{16x^2y^5\sqrt[3]{3xy^2}}$$

$$\sqrt[3]{b^{24} c^{36} d^{38}}$$

$$b^{24/3} \quad c^{36/3} \quad d^{36} \cdot d^{2}$$

$$= \boxed{b^{8} c^{12} d^{12} \sqrt[3]{d^{2}}}$$

$$\sqrt[3]{18 a^{6} b^{23} c^{14}}$$

$$c^{12/3} \cdot c^{2}$$

$$2 \quad 9$$

$$3 \quad 3$$

$$a^{6/3} \quad b^{21/3} \cdot b^{2}$$

$$\boxed{a^{2} b^{7} c^{4} \sqrt{18 b^{2} c^{2}}}$$